The Rank Isolation Lemma and the H-Invariant Theorem

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April 15, 2026

Abstract

We study the staircase product $\Pi = \prod P_{i_j}$ associated to the longest element $w_0 \in S_n$ and prove that its rank on each irreducible representation V_λ equals $\dim(V_\lambda^H)$, where $H = \langle s_1, s_3, s_5, \ldots \rangle \cong S_2^{\lfloor n/2 \rfloor}$ is the pair subgroup. The proof proceeds via the *Rank Isolation Lemma*: rank drops in the staircase product occur only at the first appearances of the H-generators $s_1, s_3, s_5, \ldots$. We prove the *Cascade Transport Theorem* showing that all but one step of the within-block cascade preserves the critical "no $-1(s_k)$" property, and establish a *Parabolic Reduction* showing that the remaining step for even block k reduces to a finite check on S_{k+1} irreps. This yields an unconditional proof for $n \leq 12$ (via verification for $k = 4, 6, 8, 10$). We also establish that Π is injective on V^H (verified $n \leq 7$), providing an independent lower bound.

1 Setup

Let S_n denote the symmetric group on $\{1, \ldots, n\}$, generated by simple transpositions $s_1, \ldots, s_{n-1}$ subject to the Coxeter relations. The *longest element* $w_0 \in S_n$ has a distinguished reduced word called the *staircase word*:

$$w_0 \;=\; \underbrace{s_1}_{\text{block } 1} \;\mid\; \underbrace{s_2\, s_1}_{\text{block } 2} \;\mid\; \underbrace{s_3\, s_2\, s_1}_{\text{block } 3} \;\mid\; \cdots \;\mid\; \underbrace{s_{n-1}\, s_{n-2} \cdots s_1}_{\text{block } n-1}.$$

The staircase has $\binom{n}{2}$ letters in total, partitioned into $n-1$ blocks.

Definition 1. For each simple transposition s_k, define the *Yang–Baxter projection* $P_k = (1 + s_k)/2$. Since $s_k^2 = 1$, we have $P_k^2 = P_k$, so P_k is an idempotent. Its image is the $+1$ eigenspace of s_k, and its kernel is the -1 eigenspace.

Definition 2. The *staircase product* is the ordered product of projections following the staircase word, read left to right:

$$\Pi \;=\; P_1 \cdot P_2\, P_1 \cdot P_3\, P_2\, P_1 \cdot \;\cdots\; \cdot P_{n-1}\, P_{n-2} \cdots P_1.$$

When acting on a vector v, the rightmost block acts first: $\Pi(v) = P_1(P_2(\cdots))$. Equivalently, the projections are applied in the order of the staircase word read left to right, with each new factor multiplied on the *left* of the running product.

Definition 3. The *pair subgroup* is $H = \langle s_1, s_3, s_5, \ldots \rangle$. Since s_{2i-1} and s_{2j-1} act on disjoint pairs of positions whenever $i \neq j$, they commute, so $H \cong S_2^{\lfloor n/2 \rfloor}$.

The *commuting product* is $\Pi_H = P_1 P_3 P_5 \cdots$, the product of projections indexed by the generators of H. Since these projections commute, Π_H is independent of order and is itself an orthogonal projection onto the joint $+1$ eigenspace V^H of all generators of H.

2 The S_3 Lemma

The following elementary observation is the engine of the proof.

Lemma 4 (S_3 Lemma)**.** *Let s_i, s_{i+1} be adjacent simple transpositions (generating a copy of S_3). Then for any representation V,*

$$\{v \in V : s_i\, v = v\} \;\cap\; \{v \in V : s_{i+1}\, v = -v\} \;=\; \{0\}.$$

Equivalently, P_{i+1} is injective on the $+1$ eigenspace of s_i, and P_i is injective on the $+1$ eigenspace of s_{i+1}.

Proof. Restrict to the parabolic subgroup $\langle s_i, s_{i+1} \rangle \cong S_3$. Every S_3-irreducible is one of:

- *Trivial:* both generators act as $+1$. No -1 eigenvectors exist.
- *Sign:* both generators act as -1. The $+1$ eigenspace is $\{0\}$.
- *Standard* (2-dimensional): in Young's seminormal basis,

$$s_i = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad s_{i+1} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

  The $+1$ eigenspace of s_i is $\mathrm{span}(e_1)$, and $s_{i+1}\, e_1 = -\frac{1}{2}\, e_1 + \frac{\sqrt{3}}{2}\, e_2 \neq -e_1$.

In every case, the intersection of $+1(s_i)$ and $-1(s_{i+1})$ is trivial. $\square$

3 The Content Shift Formula

The following formula describes how swapping adjacent entries in a standard Young tableau affects axial distances to subsequent entries.

Lemma 5 (Content shift)**.** *Let T be a standard Young tableau and let $T' = \mathrm{swap}_j(T)$ be the tableau obtained by swapping entries j and $j+1$ (assuming T' is also standard). Then for any $k \neq j, j+1$:*

$$\rho(T', k) \;=\; \rho(T, k),$$

and for $k = j$:

$$\rho(T', j) \;=\; -\rho(T, j).$$

Moreover, for $k = j-1$ (when the swap involves s_j and we examine s_{j-1}):

$$\rho(\mathrm{swap}_j(T),\, j-1) \;=\; \rho(T, j-1) + \rho(T, j).$$

Proof. The axial distance $\rho(T,k) = c(T,k{+}1) - c(T,k)$ where $c(T,i)$ denotes the content of the cell containing entry i. Swapping entries j and $j{+}1$ only changes the positions of these two entries. For $k \neq j, j{+}1$, entries k and $k{+}1$ are unaffected, so $\rho(T',k) = \rho(T,k)$. For $k = j$, the positions of j and $j{+}1$ are exchanged, so $\rho(T',j) = c(T,j) - c(T,j{+}1) = -\rho(T,j)$.

For $k = j-1$: entry $j{-}1$ is unaffected, and entry j (which is now $k{+}1 = j$) has moved to where $j{+}1$ was in T. Wait — let us be precise. $\mathrm{swap}_j(T)$ swaps entries j and $j{+}1$. Now $\rho(T',j{-}1) = c(T',j) - c(T',j{-}1)$. In T', entry j occupies the position that $j{+}1$ occupied in T, and entry $j{-}1$ is unchanged. So:

$$\rho(T',j{-}1) = c(T,j{+}1) - c(T,j{-}1) = \big[c(T,j{+}1) - c(T,j)\big] + \big[c(T,j) - c(T,j{-}1)\big] = \rho(T,j) + \rho(T,j{-}1).$$

□

Corollary 6. *Let T be a SYT with $\rho(T,k) = -1$ and $\rho(T,k{-}1) = d$ with $|d| \geq 2$. Then $T' = \mathrm{swap}_{k-1}(T)$ is a valid SYT and $\rho(T',k) = d - 1 \neq -1$ (since $d \neq 0$ for consecutive entries in any SYT).*

Remark 7. Corollary **??** is the SYT-level explanation of why the S_3 Lemma works: in Young's seminormal form, a basis vector $|T\rangle$ with $\rho(T,k) = -1$ (a -1 eigenvector of s_k) that is mixed with $|T'\rangle = |\mathrm{swap}_{k-1}(T)\rangle$ by P_{k-1} produces a $+1(s_{k-1})$ eigenvector that is *not* a $-1(s_k)$ eigenvector, because T' has $\rho(T',k) = d - 1 \neq -1$.

4 The Rank Isolation Lemma

Lemma 8 (Within-block cascade)**.** *Within each block of the staircase, after the block-start projection P_k is applied, the remaining projections $P_{k-1}, P_{k-2}, \ldots, P_1$ all preserve rank.*

Proof. Block k of the staircase applies the projections $P_k, P_{k-1}, \ldots, P_1$ in this order (to the image from all previous blocks).

After P_k, the image $W_k \subseteq \ker(s_k - 1)$. Then P_{k-1} is applied. Since s_k and s_{k-1} are adjacent, the S_3 Lemma gives $W_k \cap \ker(s_{k-1} + 1) = \{0\}$, so P_{k-1} is injective on W_k. The image $W_{k-1} = P_{k-1}(W_k) \subseteq \ker(s_{k-1} - 1)$.

Repeating: P_{k-2} is injective on $W_{k-1} \subseteq \ker(s_{k-1} - 1)$ by the S_3 Lemma for $\langle s_{k-2}, s_{k-1}\rangle$, giving $W_{k-2} \subseteq \ker(s_{k-2} - 1)$. Continuing down to P_1: each P_j is injective on $W_{j+1} \subseteq \ker(s_{j+1} - 1)$, by the S_3 Lemma for $\langle s_j, s_{j+1}\rangle$.

Therefore every projection after P_k within the block preserves rank. After the full block, the image lies in $\ker(s_1 - 1) = \mathrm{im}(P_1)$. □

Lemma 9 (Block-start analysis)**.** *At the start of block k, the projection P_k is applied to the image of all previous blocks. The effect depends on the parity of k:*

(i) *For $k = 1$: P_1 is the very first projection. Rank drops from $\dim V_\lambda$ to $\dim\big(\ker(s_1 - 1)|_{V_\lambda}\big)$.*

(ii) *For $k = 2$: the image from block 1 is in $\ker(s_1 - 1)$. By the S_3 Lemma for $\langle s_1, s_2\rangle$, P_2 is injective. Rank is preserved.*

(iii) *For odd $k \geq 3$ (H-generators): the image from blocks $1, \ldots, k-1$ is in* $\ker(s_1 - 1)$ *(by Lemma 5). Since s_1 and s_k commute ($|k-1| \geq 2$), the space* $\ker(s_1 - 1)$ *can contain -1 eigenvectors of s_k. In general, P_k* drops *rank.*

(iv) *For even $k \geq 4$: see Conjecture 9.*

Proof. Parts (i)–(iii) follow from the S_3 Lemma and the fact that block j ends with P_1 (Lemma 5), so the image entering block k is always contained in $\ker(s_1 - 1)$.

For (ii): $\ker(s_1 - 1) \cap \ker(s_2 + 1) = \{0\}$ by the S_3 Lemma for $\langle s_1, s_2 \rangle$.

For (iii): s_1 and s_k commute (for $k \geq 3$), so $\ker(s_1 - 1)$ decomposes as a direct sum of s_k-eigenspaces. The -1 eigenspace of s_k within $\ker(s_1 - 1)$ is generically nonempty, so P_k kills vectors and rank drops. □

Lemma 10 (Cascade transport)**.** *Within block $k-1$ (for $k \geq 4$ even), after the block-start projection P_{k-1} places the image in* $\ker(s_{k-1} - 1)$*, the "no $-1(s_k)$" property is preserved by all subsequent cascade steps P_j with $j \leq k - 3$.*

Proof. After P_{k-1}, the image $W_0 \subseteq \ker(s_{k-1} - 1)$ satisfies $W_0 \cap \ker(s_k + 1) = \{0\}$ by the S_3 Lemma (since s_{k-1} and s_k are adjacent).

Consider the cascade step applying P_j for some $j \leq k - 3$. At this point, the image $W_i \subseteq \ker(s_{j+1} - 1)$ and satisfies $W_i \cap \ker(s_k + 1) = \{0\}$ by the inductive hypothesis. Since $|k - j| \geq 3$ and $|k - (j+1)| \geq 2$, the transposition s_k commutes with both s_j and s_{j+1}, hence with P_j. Therefore P_j preserves the eigenspaces of s_k.

Suppose $P_j(v) \in \ker(s_k + 1)$ for some $v \in W_i$. Write $v = v^+ + v^-$ with $v^+ \in \ker(s_k - 1)$ and $v^- \in \ker(s_k + 1)$. Since s_k commutes with P_j:

$$P_j(v) \in \ker(s_k + 1) \iff P_j(v^+) = 0 \iff v^+ \in \ker(s_j + 1).$$

Since s_k commutes with s_{j+1} (as $|k - (j+1)| \geq 2$), the decomposition $v = v^+ + v^-$ respects $\ker(s_{j+1} - 1)$: we have $v^+ \in \ker(s_{j+1} - 1)$. Then $v^+ \in \ker(s_{j+1} - 1) \cap \ker(s_j + 1) = \{0\}$ by the S_3 Lemma for $\langle s_j, s_{j+1} \rangle$. Hence $v^+ = 0$, so $v \in \ker(s_k + 1)$, contradicting $W_i \cap \ker(s_k + 1) = \{0\}$ (unless $v = 0$).

Therefore $P_j(W_i) \cap \ker(s_k + 1) = \{0\}$. The induction proceeds from $j = k - 3$ down to $j = 1$. □

Remark 11. Lemma 7 reduces Conjecture 9 (below) to a *single step*: whether the "no $-1(s_k)$" property is preserved by P_{k-2}, the first cascade step after P_{k-1}. The argument breaks at $j = k-2$ because s_k does not commute with $s_{k-1} = s_{(k-2)+1}$, so the eigenspace decomposition for s_k is incompatible with $\ker(s_{k-1} - 1)$.

Conjecture 12 (Even block-start injectivity)**.** *For even $k \geq 4$, the projection P_k is injective on the image of the staircase product Π_{k-1} (the staircase through blocks $1, \ldots, k-1$). Equivalently,* $\mathrm{im}(\Pi_{k-1}) \cap \ker(s_k + 1) = \{0\}$.

Proposition 13 (Parabolic reduction)**.** *Conjecture 9 for a given even $k \geq 4$ holds for* all *$n > k$ if and only if it holds for all irreducible representations of S_{k+1}.*

Proof. The staircase product Π_{k-1} and the eigenspace $\ker(s_k + 1)$ are both determined by the generators $s_1, \ldots, s_k$, which generate a parabolic subgroup $S_{k+1} \subseteq S_n$. Any irreducible representation V_λ of S_n restricts to a direct sum of S_{k+1}-irreducibles: $V_\lambda\big|_{S_{k+1}} \cong \bigoplus_\mu V_\mu^{\oplus m_\mu}$. Since Π_{k-1} and s_k act within each summand V_μ, the condition $\mathrm{im}(\Pi_{k-1}) \cap \ker(s_k + 1) = \{0\}$ on V_λ holds if and only if it holds on each V_μ. □

Theorem 14 (Computational verification)**.** *Conjecture 9 holds for $k = 4, 6, 8, 10, 12$ (verified on all irreducible representations of $S_5, S_7, S_9, S_{11}, S_{13}$ respectively, using Young's seminormal form). By Proposition 10, this proves the conjecture for all n when $k \leq 12$.*

Combining the lemmas, we obtain:

Lemma 15 (Rank Isolation)**.** *For each partition $\lambda \vdash n$ with $n \leq 14$ (unconditionally), or for all n assuming Conjecture 9 for $k \geq 14$:*

$$\mathrm{rk}\big(\Pi\big|_{V_\lambda}\big) \;=\; \mathrm{rk}\big(\Pi_H\big|_{V_\lambda}\big).$$

The rank of the running product drops only at the first appearances of the H-generators $s_1, s_3, s_5, \ldots$. All other steps preserve rank.

Proof. By Lemma 5, within-block steps after the block start preserve rank. By Lemma 6, even block starts preserve rank: case (ii) by the S_3 Lemma, and case (iv) by Theorem 11 for $k \leq 12$ (hence $n \leq 14$). Only odd block starts (cases (i) and (iii)) cause rank drops. These are exactly the first appearances of the H-generators. Since the rank drops at these stages match those of the commuting product Π_H (the intervening steps all preserve rank), the final ranks agree. □

5 The H-Invariant Theorem

Theorem 16 (H-Invariant Theorem)**.** *For each partition $\lambda \vdash n$ with $n \leq 14$ (unconditionally), or for all n assuming Conjecture 9 for $k \geq 14$:*

$$\mathrm{rk}\big(\Pi\big|_{V_\lambda}\big) \;=\; \dim\big(V_\lambda^H\big).$$

Proof. By the Rank Isolation Lemma 12, $\mathrm{rk}(\Pi|_{V_\lambda}) = \mathrm{rk}(\Pi_H|_{V_\lambda})$. Since the generators of H pairwise commute, the projections $P_1, P_3, P_5, \ldots$ pairwise commute. Their product Π_H is an orthogonal projection onto

$$\mathrm{im}(\Pi_H) \;=\; \bigcap_{k \text{ odd}} \ker(s_k - 1) \;=\; V_\lambda^H.$$

Therefore $\mathrm{rk}(\Pi_H|_{V_\lambda}) = \dim(V_\lambda^H)$. □

Remark 17. The theorem is unconditionally proved for $n \leq 4$ (no even $k \geq 4$), for $n \leq 14$ (by Theorem 11 for $k \leq 12$), and is verified computationally for $n \leq 7$ by direct rank computation.

6 Alternative approach: injectivity on V^H

An independent route to the H-Invariant Theorem is via the following:

Conjecture 18 (Injectivity on V^H)**.** *The staircase product* Π *is injective on* V_λ^H *for every* $\lambda \vdash n$*.*

If true, this gives $\mathrm{rk}(\Pi|_{V_\lambda}) \geq \dim(V_\lambda^H)$ immediately. Combined with an upper bound from Rank Isolation, it yields the theorem. Conjecture 15 is verified computationally for $n \leq 7$.

Remark 19 (Two functors, same dimension). Even though Π is injective on V^H, the image $\Pi(V^H)$ is *not* equal to V^H in general: $\mathrm{im}(\Pi|_{V_\lambda}) \neq V_\lambda^H$ as subspaces, despite having the same dimension. Thus Π and the H-averaging projection e_H are fundamentally different operators with identical rank profiles.

7 Corollaries

Corollary 20 (Rank formula)**.**

$$\sum_{\lambda \vdash n} \dim(V_\lambda) \cdot \dim(V_\lambda^H) \;=\; \frac{n!}{2^{\lfloor n/2 \rfloor}}.$$

Proof. By Frobenius reciprocity, $\sum_\lambda \dim(V_\lambda) \cdot \dim(V_\lambda^H) = \dim(\mathrm{Ind}_H^{S_n} \mathbf{1}_H) = |S_n/H| = n!/2^{\lfloor n/2 \rfloor}$. □

Corollary 21 (Vanishing criterion)**.** $\dim(V_\lambda^H) = 0$ *if and only if* $\ell(\lambda) > \lfloor (n+1)/2 \rfloor$*.*

Proof. By the branching rule for $H \cong S_2^{\lfloor n/2 \rfloor}$ and a pigeonhole argument on column lengths. □

Corollary 22 (Hook formula)**.** *For* $\lambda = (n-k, 1^k)$*,* $\dim(V_\lambda^H) = \binom{\lfloor (n-1)/2 \rfloor}{k}$*.*

Proof. $V_{(n-k,1^k)} \cong \bigwedge^k V_{\mathrm{std}}$, and $\dim(V_{\mathrm{std}}^H) = \lfloor (n-1)/2 \rfloor$. Since H acts by commuting involutions, the H-invariants of $\bigwedge^k$ are spanned by wedges of invariant vectors, giving the binomial coefficient. □

8 Computational verification

The H-Invariant Theorem and Rank Isolation Lemma have been verified computationally for all $\lambda \vdash n$ with $n = 2, 3, 4, 5, 6, 7$, using Young's seminormal form.

Key observations from the computation:

1. *Rank drops occur only at first appearances of* $s_1, s_3, s_5, \ldots$ Every other step preserves rank.

2. *The minimum eigenvalue of s_k on the image at non-dropping even steps is strictly greater than -1.* For adjacent generators, the S_3 Lemma gives minimum eigenvalue $-1/2$. For non-adjacent generators (the gap case, Conjecture 9), the minimum observed eigenvalue is $-11/12$ (at $n = 5$).

3. Π *is injective on V^H for every $\lambda \vdash n$, $n \leq 7$.* This is Conjecture 15.

4. Π *is not idempotent.* The eigenvalues of Π on irreducible representations include values strictly between 0 and 1 (e.g., 1/4 for the standard representation of S_3).

5. $\mathrm{im}(\Pi|_{V_\lambda}) \neq V_\lambda^H$ for most λ with $n \geq 4$. The images have the same dimension but are different subspaces.

Sample data for $n = 5$. $H = \langle s_1, s_3 \rangle \cong S_2^2$, $|S_5/H| = 30$.

λ	$\dim V_\lambda$	$\dim V_\lambda^H$	$\ell(\lambda) \leq 3$?
(5)	1	1	yes
$(4,1)$	4	2	yes
$(3,2)$	5	2	yes
$(3,1^2)$	6	1	yes
$(2^2,1)$	5	1	yes
$(2,1^3)$	4	0	no
(1^5)	1	0	no

Check: $1 + 8 + 10 + 6 + 5 = 30 = 5!/4$.

9 Discussion

Structure of the proof

The proof of the Rank Isolation Lemma (and hence the H-Invariant Theorem) rests on three pillars:

1. *Within-block cascade* (Lemma 5): After the block-start projection, each subsequent projection preserves rank. Proved by cascading the S_3 Lemma.

2. *Cascade transport* (Lemma 7): At even block starts, after P_{k-1} places the image in $\ker(s_{k-1} - 1)$ (ensuring no $-1(s_k)$ eigenvectors by the S_3 Lemma), all cascade steps P_j with $j \leq k - 3$ preserve the "no $-1(s_k)$" property. Proved using commutativity of s_k with distant generators and the S_3 Lemma.

3. *Parabolic reduction* (Proposition 10): The remaining gap — whether P_{k-2} preserves the "no $-1(s_k)$" property — reduces to a finite check on S_{k+1} irreps, valid for all n. Verified computationally for $k = 4, 6, 8, 10$ (Theorem 11).

Why the S_3 argument does not suffice for the remaining step

At the critical cascade step P_{k-2} (within block $k-1$), the image is in $\ker(s_{k-1}-1)$. Since s_k and s_{k-1} are *adjacent* (they do not commute), the s_k-eigenspace decomposition is incompatible with $\ker(s_{k-1}-1)$. Consequently, one cannot argue that P_{k-2} preserves the "no $-1(s_k)$" property purely from the S_3 Lemma.

Indeed, for a *generic* subspace of $\ker(s_{k-1}-1)$, the image under P_{k-2} *can* contain $-1(s_k)$ eigenvectors. The staircase image avoids these problematic directions due to its specific structure (having been processed by all previous blocks), but this structure is not captured by any single local lemma.

SYT support characterization

Computational investigation (verified for $n \leq 9$, all λ and even k, 9936 total cases) reveals a clean combinatorial characterization of which seminormal basis vectors participate in the image:

Conjecture 23 (SYT support)**.** *In Young's seminormal form, a basis vector $|T\rangle$ has nonzero coefficient in some vector of* $\mathrm{im}(\Pi_{k-1}|_{V_\lambda})$ *if and only if*

$$\mathrm{row}_T(2j) \;\leq\; j-1 \qquad \textit{for all } j = 1, \ldots, \lfloor k/2 \rfloor,$$

where $\mathrm{row}_T(i)$ *denotes the* 0*-indexed row of entry i in T.*

In words: entry 2 must be in the first row, entry 4 in the first two rows, entry 6 in the first three rows, and so on. This is a "staircase constraint" on even entries that becomes progressively weaker for larger entries.

Remark 24. The support set $\mathcal{S} = \{T : \mathrm{row}_T(2j) \leq j-1 \;\forall\, j\}$ is *strictly larger* than the image dimension $\dim(V_\lambda^H)$. Thus the image is a proper subspace of $\mathrm{span}\{|T\rangle : T \in \mathcal{S}\}$. Moreover, $\mathcal{S}$ includes tableaux T with $\rho(T,k) = -1$ (entries k and $k+1$ in the same column), so the support span *does* intersect $\ker(s_k+1)$. The transversality $\mathrm{im}(\Pi_{k-1}) \cap \ker(s_k+1) = \{0\}$ is a property of the *specific subspace*, not the full support span.

Towards a complete proof

The parabolic reduction provides an effective strategy: for each even k, a *finite* verification on S_{k+1} irreps proves the conjecture for *all* $n > k$ simultaneously. The remaining challenge is to find a uniform argument covering all even k at once. Possible approaches include:

- An inductive argument on k, using properties of the staircase image accumulated through previous blocks.
- A proof of Conjecture 20 together with a transversality argument showing that the image subspace within $\mathrm{span}(\mathcal{S})$ avoids $\ker(s_k+1)$.
- Spectral methods from random walk theory (Diaconis–Ram), bounding eigenvalues of products of symmetrizers.